WORKSHEET 7.3

Drawing Slope Fields

Unit 7: Differential Equations

Key ideas

- A **slope field** shows, at each point (x, y), a short segment whose slope is the value of $\dfrac{dy}{dx}$ there.
- To draw one, substitute each point into the differential equation.
- Segments are horizontal wherever $\dfrac{dy}{dx} = 0$.
- If $\dfrac{dy}{dx}$ depends only on x, every column of the field is identical; if only on y, every row is.

Worked examples

Example 1 · Computing slopes

For $\dfrac{dy}{dx} = y - 2x$, find the slopes at $(1, 1)$, $(0, 2)$ and $(2, 4)$.

Solution. Substitute each point.

-1, 2 and 0.

Example 2 · Horizontal segments

Where are the segments of the slope field for $\dfrac{dy}{dx} = y - 2x$ horizontal?

Solution. Set $y - 2x = 0$.

Along the line $y = 2x$.

Example 3 · Recognising a pattern

A slope field has the same segments all the way up each vertical line. What does this say about the differential equation?

Solution. The slope does not change when y changes.

$\dfrac{dy}{dx}$ depends on x only.

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Practice

CORE

1. For $\frac{dy}{dx} = x^2y$, find the slope at $(2, 3)$.

2. For $\frac{dy}{dx} = x^2y$, find the slope at $(-1, -2)$.

3. For $\frac{dy}{dx} = x^2y$, where are the segments horizontal?

4. For $\frac{dy}{dx} = x^2y$, what is the sign of the slopes in the third quadrant?

EXAM LEVEL

5. For $\frac{dy}{dx} = x^2 + y^2$, explain why no segment has negative slope, and find where the slope is zero.

6. Describe the slope field of $\frac{dy}{dx} = (y - 1)(y + 2)$: where are the segments horizontal, and what is special about each row?

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7. Describe the slope field of $\dfrac{dy}{dx} = x - 2$, and the shape of its solution curves.

8. A slope field has horizontal segments along the curve $y = x^3$, positive slopes above that curve and negative slopes below it. Suggest a differential equation.

Question 9 · Free-response practice

Consider the differential equation $\dfrac{dy}{dx} = y(2 - x)$.

(a) Find the slope at each of the points $(0, 1)$, $(2, 3)$, $(3, 2)$ and $(1, -1)$.

(b) Describe all points at which the segments of the slope field are horizontal.

(c) The solution through $(2, 3)$ has a horizontal tangent there. Is this a relative maximum or a relative minimum? Justify your answer.

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Worked solutions

1. 12.
2. -2.
3. Along both axes: $x = 0$ or $y = 0$.
4. Negative ($x^2 > 0$, $y < 0$).
5. $x^2 + y^2 \geq 0$ always; it is zero only at the origin.
6. Horizontal along $y = 1$ and $y = -2$. The slope depends only on y, so each row has identical segments. Slopes are negative between the two lines and positive outside them.
7. The slope depends only on x: columns are identical, negative for $x < 2$ and positive for $x > 2$. Solutions are parabolas $y = \dfrac{x^2}{2} - 2x + C$ with their lowest points at $x = 2$.
8. $\dfrac{dy}{dx} = y - x^3$.
9. (a) 2, 0, -2 and -1.
   (b) The line $y = 0$ and the line $x = 2$.
   (c) $\dfrac{d^2y}{dx^2} = \dfrac{dy}{dx}(2 - x) - y$, which equals -3 at $(2, 3)$. A zero first derivative and a negative second derivative: a relative maximum.

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